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PRENATAL SOUND STIMULATION ON POSTNATAL RAT OFFSPRING OPEN FIELD BEHAVIORS[1]

MIKE MORRA
Xavier University

The impact of an intrinsically neutral stimulus (standard door buzzer), presented in a novel setting during the second semester of pregnancy, was capable of altering rat offspring behavior in the open field on Day 25, postpartum. Twenty-four female albino rats of approximately 120 days of age were used. This sound-stressed group was compared to two control groups; a conditioned-fear group and a non-stressed group. The direction of the behavioral differences was increased emotionality between the sound-fear group and the control group ($p < .01$). These findings suggested that the stressor may have acted directly on the fetus rather than indirectly via the placental barrier.

The adverse effects of maternal anxiety during pregnancy on offspring behaviors have been reported by a number of investigators. These studies presented evidence of significant increase of emotionality (Ader & Belfer, 1962; Doyle & Yule, 1959; Hockman, 1961; Morra, 1965; Thompson, Watson & Charlesworth, 1962), reduced cognitive efficiency (Davids, Holden & Gray, 1963; Graham, Ernhart, Thurston & Craft, 1962), and decreased vitality and viability (Antonov, 1947; Hockman, 1961; Morra, 1965). On the other hand, "positive effects" (greater resistance to stress; less emotionality) have been reported by Ader and Conklin (1963), with prenatal handling; Furchtgott and Echols (1958), with prenatal x-irradiation; and Vincent (1958), with prenatal alcoholism.

These effects of maternal anxiety have been attributed to conditioned avoidance (Thompson, 1957) and to the direct action of a maternal noxious experience (Antonov, 1947; Sontag, 1944). However, contemporary motivation theory suggests that the impact of *any* stimulus on an organism is derived not only from its inherent capability to produce approach and avoidance behaviors (i.e. meaningfulness, but also from its characteristics of intensity and variation. (Fiske & Maddi, 1961, pp. 11-56). Consequently, the impetus of a "neutral" stimulus (e.g. standard door buzzer) should be capable of inducing fear when it causes the organism to depart markedly from optimal levels of arousal.

This study attempted to test this proposition in a design in which a moderate level of intermittent sound stimulation was administered to female albino rats, in a novel setting, during the second semester

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of pregnancy. The behaviors of these animals were compared to those of two control groups, a pre-pregnancy avoidance trained group and a handled-only group. The avoidance trained group was used as a control group in that all previous prenatal stress studies, to date, have considered avoidance conditioning in their designs.

METHOD

Subjects

Twenty-four female rats of Wistar strain, obtained from the Budd Mountain Rodent Farm, were divided randomly into three groups. These animals were between 120 and 140 days of age at the beginning of the experiment. In addition, 18 male albino rats, approximately 150 days of age, were used in the mating procedure. These animals were assigned randomly to six cages in groups of three rats per cage.

Of the female rats, Group I received pre-pregnancy avoidance training and exposure to the conditioned stimulus during the second phase of pregnancy. Group II received only the CS exposure treatment during the same second period of pregnancy and Group III received only equal handling during the same period during pregnancy. Thirteen out of the 24 mothers produced litters (six in Group III; three in Group II; and four in Group I). These litters were all born within 6 days, i.e., Days 22-27 (counting from Day 1 of mating).

Sixty-three offspring from Group III, 35 offspring from Group II, and 26 offspring from Group I were tested in the open field. The sizes of the litters on Day 25 are reported in Table 1. These differences in viability were consistent with the differences reported by other investigators (Ader & Belfer, 1962; Doyle & Yule, 1959; Hockman, 1961; Morra, 1965). An analysis of the fertility and viability data was not attempted since it has already been demonstrated that these differences were not due to specific experimental conditions. The actual variables associated with these findings have not yet been researched.

Apparatus

The apparatus used in the avoidance training was a modified Mowrer-Miller shock avoidance box having interior dimensions as follows: length, 36 in.; width, 5 in.; height, 2½ ft. The grid floor consisted of 1/16-in. diameter bars spaced ⅜ in. apart horizontally, and beneath this grid there were four bars equally spaced running lengthwise. A 3-in. high electrified barrier across the middle of the box divided it into two parts. The CS was a noise of 3.25 sec. duration produced by a door buzzer. The UCS was a 3.0 ma electric shock to the grids from a dc constant current generator which came on .35 sec. after the cessation of the CS. An electric timer was activated at the time that the CS was terminated. Exactly one minute after the cessation of the CS, a new trial was begun with shock now administered to the other side. A commutator motor scrambled the grid bars so that all bars were charged in relation to each other. The entire apparatus, including the recycling, was automatic so that *E* had only to record the avoidance behaviors. The estimated over-all sound pressure in db (Re: .0002 microbar) was 85.

The open field apparatus used in the offspring testing was constructed out of ½-in. plywood and was unpainted. Its dimensions were 25 in. long, 25 in. wide, 2 ft. high. Twenty-five squares were marked off in red paint and uniform lighting was provided by an overhanging 60 w bulb.

Procedure

Each animal in Group I was placed in the non-shock side of the shuttlebox at the beginning of the trial phase. It received avoidance training at the rate of 20 trials a day for six consecutive days. At the end of this pretraining, the animals of all three groups were placed in mating cages for three consecutive days. Each mating cage consisted of four females and three males. This mating procedure was used in order to minimize the effects of handling during pregnancy and to avoid the possible confounding effects of the additional stress brought about by the vaginal smear technique.

On Day 12 (counting from Day 1 of mating), the animals in Groups I and II were returned to the avoidance apparatus and were exposed only to the sound of the buzzer (i.e. CS for Group I) for eight consecutive days. Each animal received three trials, twice daily, with an inter-block interval of approximately one hour. The animals were not prevented from traversing to the adjacent section of the shuttlebox. Group III received equal and similar handling throughout this same period. This handling included the removal of the animal from its home cage, the placing of it into the avoidance apparatus, and the return to the home cage.

On Day 19 (again counting from Day 1 of mating), the animals were isolated and were given equal amounts of nesting materials. No cross-fostering was attempted since no primary effects of fostering were reported by other investigators. On Day 23 postpartum, the offspring were weaned and 2 days later they were tested in the open field for one one-minute trial per rat.

RESULTS AND DISCUSSION

The data for the offspring are presented in Table 1. Analysis of variance tests were applied to measures of frequency of traversals and initial latency times. Orthogonol comparisons of the groups were not necessary since the differences obtained were obvious upon inspection of the data. The analysis of the traversal measure yielded differences between the control group III and both stress groups ($F=5.56$; $df=2/123$; $p<.01$). The stressed groups were characterized by less activity. The analysis of the latency measure yielded significant differences between Group II (sound-stressed) and Groups I and III ($F=5.84$; $df=2/123$; $p<.01$). With both measures, the sound-stressed group differed significantly from the control group (III) in the direction of greater emotionality (i.e. less activity and greater latency in a novel setting). The number of male and female offspring among the three groups were not significantly different, consequently no analysis of the sex variable was necessary. Ader and Belfer (1962) found that ". . . irrespective of sex . . ., the offspring of the manipulated mothers were significantly more emotional."

TABLE 1
OPEN FIELD BEHAVIORS OF RAT OFFSPRING ON DAY 25

	Conditioned Fear (Group I)	Sound Fear (Group II)	Handled Only (Group III)
Number Offspring	26	35	63
Mean Number Traversals	15.8	14.2	19.5
σ Traversals	5.7	8.3	8.4
Mean Initial Latencies (sec)	13.5	18.8	13.1
σ Latencies (sec)	6.1	8.2	7.9

The proposition tested in this study was supported by the data. That is, the impact of an intrinsically neutral stimulus (standard door buzzer) presented in a novel setting, during the second semester of pregnancy, was capable of altering rat offspring behaviors on Day 25, postpartum. The direction of the behavioral differences was increased emotionality between the sound-stressed group (II) and the control group. The conditioned-fear group did not differ from the sound-stressed group in activity which suggested that the findings reported by Thompson (1957; 1962) were confounded by the systematic effects of the buzzer treatment during pregnancy. It is possible that the fear-component produced by this novel experience accounted for a considerable portion of the variation reported by Thompson (1957).

Interestingly enough, the question of whether these offspring behavioral differences were due to the stress state of the maternal host remains unresolved. The fact that a smaller number of litters occurred among the mothers of Groups I and II (stress groups) might be taken to support the notion of basic disturbances in maternal physiology. Variations in the productivity of the experimental litters as a result of special conditions during pregnancy has also been noted by other investigators (e.g. Hockman, 1961). It is possible that certain types of "psychological" stressors are capable of affecting the fetus directly rather than indirectly via the placental barrier. Currently, a replication of this study is in process in which fertile chick eggs are being used as Ss. This choice of organism will tend to factor out the effects of the maternal influences on the developing fetus.

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